3.3.2 Verlet

Recall the position and velocity Verlet methods:

$$\begin{aligned} &\textbf{Position Verlet}: && \textbf{Velocity Verlet}: \\ &\hat{\boldsymbol{Q}} := \boldsymbol{q} + (h/2)\, \boldsymbol{M}^{-1}\boldsymbol{p}, && \hat{\boldsymbol{P}} := \boldsymbol{p} - (h/2)\, \nabla U(\boldsymbol{q}), \\ &\boldsymbol{P} := \boldsymbol{p} - h\, \nabla U(\hat{\boldsymbol{Q}}), && \boldsymbol{Q} := \boldsymbol{q} + h\boldsymbol{M}^{-1}\hat{\boldsymbol{P}}, \\ &\boldsymbol{Q} := \hat{\boldsymbol{Q}} + (h/2)\, \boldsymbol{M}^{-1}\boldsymbol{P}. && \boldsymbol{P} := \hat{\boldsymbol{P}} - (h/2)\, \nabla U(\boldsymbol{Q}). \end{aligned} \tag{3.3}$$

We can think of the velocity Verlet method as being defined by a splitting into three parts:

$$\begin{aligned} H(\boldsymbol{q},\boldsymbol{p}) &= H_1 + H_2 + H_3, \\ H_1 &= \frac{1}{2}U(\boldsymbol{q}), \\ H_2 &= \frac{1}{2}\boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p}, \\ H_3 &= \frac{1}{2}U(\boldsymbol{q}). \end{aligned}$$

This is a sequence of two types of operations: it consists of a "kick" exhibited by a jump in the momentum, linear "drift" with the resulting momentum, followed by a final kick. The symmetry of the method is one of its important features. Switching the order of the operations, i.e. drift-kick-drift, gives the position Verlet method.

In general for the Verlet method applied to Hamiltonians of the form $H(\boldsymbol{q},\boldsymbol{p}) = T(\boldsymbol{p}) + U(\boldsymbol{q})$, the BCH lemma gives

$$\begin{aligned} \tilde{H}_h = T + U &+ \frac{h^2}{12}\left(\{T,\{T,U\}\} - \frac{1}{2}\{U,\{U,T\}\}\right) \\ &+ \frac{h^4}{120}\left(-\frac{1}{6}\{T,\{T,\{T,\{T,U\}\}\}\} + \frac{1}{3}\{U,\{T,\{T,\{T,U\}\}\}\}\right. \\ &\left. -\frac{1}{4}\{U,\{U,\{T,\{T,U\}\}\}\} + \{T,\{T,\{U,\{U,T\}\}\}\}\right) + \mathcal{O}(h^6) \end{aligned}$$

The extra power of h in the leading perturbation reflects the fact that this method is 2nd order accurate, rather than first order, as Symplectic Euler. Note that only even order terms $(h^2, h^4, \ldots)$ appear in the perturbative expansion; this is a consequence of the symmetry of the method.

We can exploit certain features of the BCH series to obtain more general second order symplectic methods. For Hamiltonians X and Y, consider symmetrizing a composition of exponentials (a Strang splitting), such that

$$\begin{aligned}\exp\left(\frac{t}{2}X\right)\exp\left(tY\right)\exp\left(\frac{t}{2}X\right) &= \exp(tZ_t)\\ &= \exp(t\tilde{Z}_{[1]} + t^2\tilde{Z}_{[2]} + t^3\tilde{Z}_{[3]} + t^4\tilde{Z}_{[4]} + \ldots).\end{aligned}$$

Multiplying by the same expansion using a different time step s, we have

$$\begin{aligned}&\exp\left(\frac{s}{2}X\right)\exp\left(sY\right)\exp\left(\frac{s}{2}X\right)\exp\left(\frac{t}{2}X\right)\exp\left(tY\right)\exp\left(\frac{t}{2}X\right)\\ &\quad= \exp(sZ_s)\exp(tZ_t).\end{aligned} \tag{3.4}$$

But we know that Z_t commutes with Z_s, giving

$$\begin{aligned}\exp\left(sZ_s\right)\exp\left(tZ_t\right) &= \exp\left(sZ_s + tZ_t\right)\\ &= \exp\left((s+t)\tilde{Z}_{[1]} + (s^2+t^2)\tilde{Z}_{[2]} + (s^3+t^3)\tilde{Z}_{[3]}\right.\\ &\quad\left.+(s^4+t^4)\tilde{Z}_{[4]} + \ldots\right).\end{aligned} \tag{3.5}$$

Setting $s = -t$, the left hand side of (3.4) collapses to the identity. Hence the surviving even-powered terms in the expansion on the right hand side of (3.5) must be zero. If the even order terms vanish, then the effect of composing linear operators symmetrically is to keep only odd order terms in the exponent, giving the symmetric BCH formula

$$\exp\left(\frac{t}{2}X\right)\exp\left(tY\right)\exp\left(\frac{t}{2}X\right) = \exp(t\left(X+Y\right) + t^3\tilde{Z}_{[3]} + \ldots), \tag{3.6}$$

where

$$\tilde{Z}_{[3]} = \frac{1}{12}[Y,[Y,X]] - \frac{1}{24}[X,[X,Y]]. \tag{3.7}$$

More complicated symmetric second-order schemes can be devised by proceeding almost arbitrarily and maintaining a symmetric composition, but it is not found that alternative approaches improve on the two Verlet schemes, at least for molecular applications. The Verlet methods are seen as the gold-standard for molecular dynamics computations: both require only one evaluation of $\nabla U(\boldsymbol{q})$ per iteration (where the velocity Verlet scheme can reuse $\nabla U(\boldsymbol{Q})$ for the next iteration), and offer a second-order symplectic evolution.